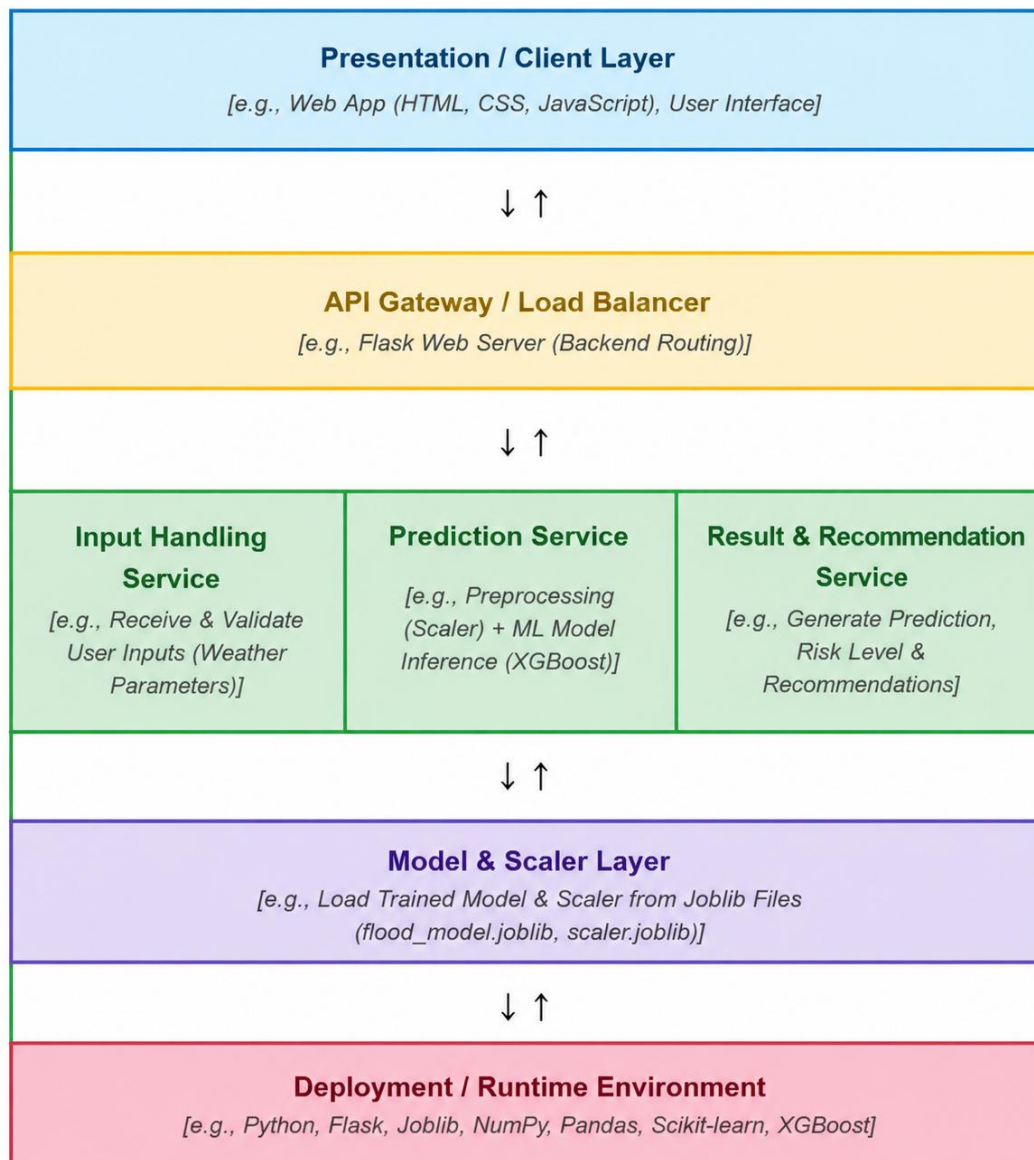


SOLUTION ARCHITECTURE

Date	27 June 2026
Team ID	
Project Name	Rising waters
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation / Client Layer	Provides the user interface where users enter weather and rainfall parameters and view flood prediction results.	HTML, CSS Templates
Flask Backend / API Layer	Handles user requests, validates inputs, loads the trained model, processes predictions, and sends the results back to the frontend.	Python, Flask
Machine Learning Service	Performs data preprocessing using the saved StandardScaler and generates flood predictions using the trained XGBoost model.	XGBoost, Scikit-learn, Joblib, NumPy
Model Repository	Stores the trained Machine Learning model (flood_model.joblib) and StandardScaler (scaler.joblib) used during prediction.	Joblib Files, Local Storage